

Explainable machine learning-assisted exploration of chromatin dynamics reveals chromosome-specific response to serum starvation

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eLife Assessment

This interesting study adapts machine learning tools to analyze movements of a chromatin locus in living cells in response to serum starvation. The machine learning approach developed is **useful**, the experiments are well controlled, and the data are **solid**. The study would be greatly strengthened by testing key predictions made using perturbation experiments. This work will be of interest to those studying chromosome biology and gene expression patterns.

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Abstract

Chromatin is dynamic at all length scales, influencing chromatin-based processes, such as gene expression. Even large-scale reorganization of whole chromosome territories has been reported upon specific signals, but lack of suitable methods has prevented analysis of the underlying dynamic processes. Here we have used CRISPR-Sirius for time-lapse imaging of chromatin loci dynamics during serum starvation. We show that the chromosome 1 loci move towards the nuclear envelope during the first hour of serum starvation in a chromosome-specific manner. Machine learning-assisted exploration of acquired multiparametric data combined with the Shapley values-based explanation approach allowed us to uncover the critical features that characterize chromatin dynamics during serum starvation. This analysis reveals that although serum starvation affects overall nuclear morphology and chromatin dynamics, chromosome 1 loci display a specific response that is characterized by maintenance of dynamics in constrained environment, and long “jumps” at the nuclear periphery. Interestingly, the two homologous chromosomes display differential behaviors, with the more peripheral homolog being more responsive to the signal than the internal one. Overall, the presented machine learning-assisted dataset exploration helps us navigate the multidimensional data to understand the underlying dynamic processes and can be applied to a wide variety of research questions in imaging and cell biology in general.

Introduction

The human genome is non-randomly organized within the cell nucleus. This complex, multilevel organization from wrapping the DNA around histones as nucleosomes to chromatin loops and topologically associated domains (TADs) and finally to chromosome territories ensures genome functionality at all stages of the cell cycle and in response to intrinsic and extrinsic cues. In addition to folding, also the localization within the 3D nucleus influences chromatin functions. Radial organization, the relative proximity to the generally repressive nuclear envelope, is one measure that can be used to describe the nuclear localization of specific chromatin loci and has been shown in many examples to correlate with not only gene activity, but also for example replication timing. In addition to the nuclear envelope, also proximity to nuclear bodies, such as nucleolus and nuclear speckles, can influence chromatin organization and functions (Bizhanova and Kaufman, 2020 [↗](#)).

An essential feature of chromatin organization, at all organizational levels, is its dynamic nature. The polymer nature and the different interactions of chromatin result in subdiffusive behavior that is more restricted than Brownian motion (Dion and Gasser, 2013 [↗](#)). However, observations of specific loci switching to what could be described as active motion were reported as early as the late 90s (Bornfleth *et al.*, 1999 [↗](#)). Subsequent transgene-based approaches revealed long-range repositioning of specific chromosome loci in response to transcriptional activation (Chuang *et al.*, 2006 [↗](#); Dundr *et al.*, 2007 [↗](#); Khanna *et al.*, 2014 [↗](#)). Interestingly, interfering with actin and myosin activity prevented repositioning of the loci. Actin machinery has also been linked to chromatin dynamics especially during DNA damage response by moving damaged DNA to favorable repair sites (Caridi *et al.*, 2018 [↗](#); Schrank *et al.*, 2018 [↗](#)). Thus, while chromatin generally displays subdiffusive dynamics, also long-range, actin-dependent movement has been described.

The particular position of each chromosome remains relatively stable during interphase, forming what is known as a chromosome territory (Walter *et al.*, 2003 [↗](#)). Nevertheless, studies using mainly fluorescent in situ hybridization (FISH) techniques have revealed large-scale repositioning of chromosome territories in response to serum removal (Mehta *et al.*, 2010 [↗](#)), DNA damage (Mehta *et al.*, 2013 [↗](#); Kulashreshtha *et al.*, 2016 [↗](#)) and senescence (Mehta *et al.*, 2021 [↗](#)), as well as during various disease and differentiation contexts (Williams *et al.*, 2002 [↗](#); Matarazzo *et al.*, 2007 [↗](#)). In primary human fibroblasts, serum withdrawal results in chromosome repositioning of about half of the chromosomes in a process that requires energy and the activity of an actomyosin motor. The repositioning is rather rapid, taking place in about 15 minutes, as well as reversible (Mehta *et al.*, 2010 [↗](#)). Since these studies have been performed in fixed cells, the underlying dynamic processes have remained unclear.

Although both FISH- and transgene-based approaches have advanced chromatin dynamics studies, they still retain several key disadvantages. FISH-based techniques do not allow following the cells live, thus losing the data on real-time motility. In turn, the transgene-based systems require the introduction of an engineered construct into the genome. This restricts the ability to observe the dynamics in untouched cells in a non-invasive way and leaves no freedom to choose the endogenous locus for study. CRISPR-based approaches utilizing endonuclease deficient Cas9 (Chen *et al.*, 2013 [↗](#); Ma *et al.*, 2016 [↗](#)) effectively address both issues, providing precise non-invasive localization of a sequence-of-choice in live cells. Indeed, for example CRISPR-Sirius system, where an engineered guide RNA (gRNA) scaffold amplifies the signal for live imaging (Ma *et al.*, 2018 [↗](#)) has revealed cell-cycle dependent chromatin dynamics (Ma *et al.*, 2019 [↗](#)). Live imaging of chromatin dynamics results in multidimensional data that has the potential to reveal mechanistic insights into the underlying biological process. Nevertheless, many previous studies concentrate

on a limited set of dynamic properties, such as mean squared displacement (MSD)-related metrics (Dion and Gasser, 2013 [↗](#)), necessitating the development of data analysis tools to fully capitalize on the information-rich imaging data.

In this study, we combined CRISPR-Sirius for sequence-specific live visualization of chromatin with machine learning-assisted dataset exploration. Explanation of the trained classifiers' decisions using a game theory-rooted approach, named SHAP (Lundberg and Lee, 2017 [↗](#)), allowed us to extract the patterns underlying chromosome 1 dynamics upon serum starvation. This manifests as fast spontaneous jumps rather than continuous movement in a particular direction. In parallel, serum starvation induces a global change in morphological characteristic of the nucleus, such as decrease in the area, which, however, do not explain the radial reorganization of chromosome 1. Finally, under starvation, homologous chromosomes exhibit differential behavior, with peripheral homologs often moving faster than central ones. We are confident that, together with the emerging advanced techniques tackling the complexity of particle tracking (Pinholt *et al.*, 2021 [↗](#)) and other multimodal biological data (Sturma *et al.*, 2023 [↗](#)), explainable ML-assisted dataset exploration will add to a toolbox allowing an information-efficient and unbiased data analysis applicable far beyond the showcased scenario.

Results

Monitoring chromosome relocation upon serum starvation with CRISPR-Sirius

Relocalization of chromosomes has previously been described upon DNA damage, heat shock and serum starvation (Abdel-Halim *et al.*, 2006 [↗](#); Mehta *et al.*, 2010 [↗](#); Mehta *et al.*, 2013 [↗](#); Kulashreshtha *et al.*, 2016 [↗](#)). These studies have been carried out in fixed cells, precluding analysis of the dynamic processes involved. To tackle this issue, we have used CRISPR-Sirius, a tool enabling live imaging of specific chromatin loci (Ma *et al.*, 2018 [↗](#)), to study chromosome relocalization upon serum starvation. To establish the feasibility of this approach, we first performed a prescreening in fixed U2OS cells that were grown in serum containing media (10%) or serum starved (0.3%) for 1h. We focused on the set of chromosomes previously suggested to display differential radial position between proliferating and quiescent nuclei of human dermal fibroblasts or upon serum starvation (Mehta *et al.*, 2010 [↗](#)), namely chromosomes 1, 10, and 13, as well as X as a control. Of the imaged chromosomes, chromosome 1 displayed detectable relocalization towards the nuclear periphery, manifested also as change in the skewness of distance to edge distribution (**Fig. 1a,b** [↗](#)). Relocalization was less evident for the rest of the screened chromosomes. To assess the relocalization of chromosome 1 similarly as in the conventional FISH-based assay (Mehta *et al.*, 2010 [↗](#); Mehta *et al.*, 2021 [↗](#)), we also analyzed its nuclear localization by dividing the nucleus into concentric shells with an equal area. Also this approach confirmed the relocalization of chromosome 1 towards the nuclear periphery (**Fig. 1c** [↗](#)). The inability to detect changes in radial positioning of the other chromosomes may be due to the differential labeling process or the used cell line. Chromosome paints used in the FISH assays cover large parts of the chromosome, whereas CRISPR-Sirius targets only specific repetitive loci within the chromosome. Moreover, here we use the osteosarcoma cell line U2OS, whereas the original studies employed human dermal fibroblasts, and it is possible that different cell lines display distinct chromatin-based responses to serum starvation. Nevertheless, the pre-screening tests showed that CRISPR-Sirius-based tools can be used to study relocalization of chromosome 1 upon serum starvation, opening the possibility for dynamic studies in living cells.

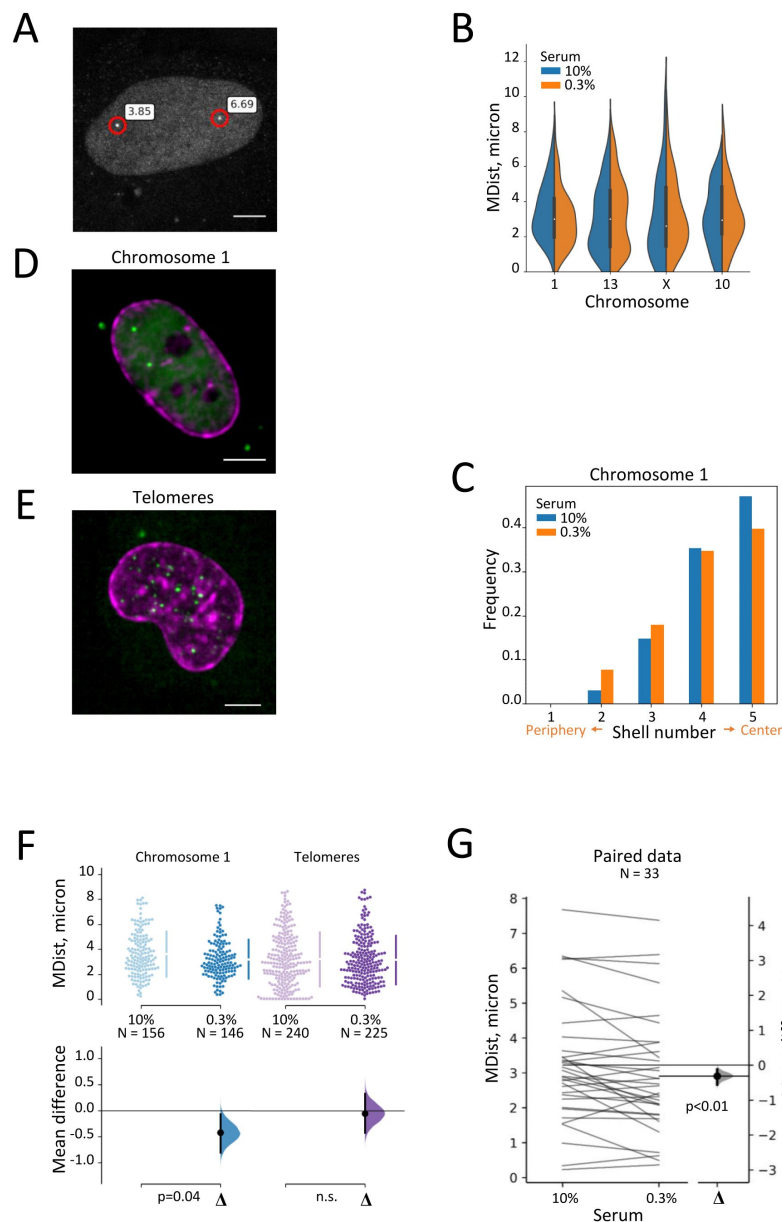


Figure 1

Chromatin dynamics upon serum starvation with CRISPR-Sirius.

a, Representative image of a nucleus in fixed sample with chromosome loci visualized using CRISPR-Sirius. Minimal distance to edge indicated in white boxes, microns. Scale bar 6 microns. **b**, Kernel density estimate distributions showing the minimal distance to the nuclear edge for different chromosome loci in serum (10%) or after 1h serum starvation (0.3%). **c**, Fractions of chromosome 1 loci detected in shells of equal volume derived from the division of the nuclear area by sequential morphological erosion. **d**, **e** Representative images of U2OS cells expressing CRISPR-Sirius plasmids with the gRNA targeting chromosome 1 (**d**) and telomeres (**e**) in green, and LaminA-mCherry for nucleus segmentation in magenta. **f**, Minimal distance to edge of visualized loci in serum (10%) and under starvation (0.3%). Each dot in a swarmplot, upper panel, represents individual locus. Lower panel represents the distributions of bootstrapped delta mean values for datasets. Error bars show 95% confidence intervals in both upper and lower panels. Statistical significance with Mann-Whitney U rank test. **g**, Chromosome 1 loci are observed to localize closer to the nuclear periphery under starvation, also when measured in the same cells before and after (within 40 min) medium change. Every line represents the change in minimal distance to the nuclear periphery for a particular locus; left scale. The distribution shows bootstrapped delta mean values for paired dataset; right scale. Error bars show 95% confidence intervals. The difference is significant, paired two-tailed t-test, normality Shapiro-Wilk ($\alpha=0.05$), $p<0.01$.

Live imaging confirms locus-specificity of chromosome 1 radial relocation upon starvation

To analyze the dynamic behavior of chromosome 1 loci in response to serum starvation, the cells were transfected with CRISPR-Sirius and LaminA-mCherry for nuclear segmentation ([Fig. 1d](#)). Randomly chosen cells were imaged in 10% FBS and in low serum (0.3% FBS), within the first 40 min into starvation. The minimum distance to the edge (MDist; all measured parameters and corresponding naming are summarized in Table 1) was calculated for each frame. As a control, we monitored the movement of telomeres ([Fig. 1e](#)) in order to distinguish locus-specific effects of serum starvation from general nucleus-level rearrangements. For chromosome 1, ~ 0.5 micron shift towards nuclear periphery was observed in live cells under serum starvation compared to cells grown in serum ([Fig. 1f](#)). While telomeres were generally more variable in terms of their radial distribution, their averaged minimal distance to the nuclear edge did not show any significant change under starvation, with the differences of the means being close to zero ([Fig. 1f](#)). To exclude possible sampling bias before and after medium change, we performed a set of experiments with the same cells imaged before and after changing the media to low serum ([Fig. 1g](#), [Supplementary Figure S1a,c](#)). Here, a statistically significant shift in mean distance (MDist, [Table 1](#)) of 0.32 microns towards the periphery was observed in starvation ([Fig. 1g](#)). When examining only the first twenty minutes after medium change, the effect was even more pronounced in this paired dataset ([Supplementary Figure S1b](#)). Thus, chromosome 1 relocates towards the nuclear periphery upon changing the media to contain low serum. This reorganization is specific to chromosome 1 and is not detectable, when analyzing telomeres.

Machine learning augmented analysis of timelapse-derived data

To fully capitalize on this methodology that allows for the real-time monitoring of loci position and nuclear morphology in live cells, we have further enhanced the acquired dataset by incorporating both manually engineered features derived from tracking, as well as the raw morphology data ([Table 1](#)). Deciphering the mechanisms underlying a particular biological process includes the choice of descriptive features of the process in question, extracting the patterns in features and their interactions and, finally, interpretation of these patterns. In biological imaging these steps usually imply data pre-processing, semantic segmentation, tracking, and modeling ([Cohen, 2017](#)). Naturally, each of these steps leads to the generation of new higher-level features. This feature space expansion, boosted by addition of complex features, derived from two or more naïve system characteristics, leads to inadequacy of human interpretation due to combinatorial explosion ([Arganda Carreras and Andrey, 2017](#); [Berisha et al., 2021](#)) and multiple comparisons problem ([Curran-Everett, 2000](#)). To mitigate this problem while maximizing the information gain, we built our analysis pipeline around machine learning-assisted dataset exploration ([Fig. 2a](#)). For this, we designed a set of machine learning models and trained them on features extracted from chromosome 1 live imaging dataset ([Table 1](#)). A gradient boosting classifier (GBC) and a multilayer perceptron (MLP), relatively simple models working robustly on tabular datasets of this size, were trained on all the features aggregated over time, including the engineered ones. Alternatively, InceptionTime, a deep-learning model architecture designed for time-series classification, was chosen to analyze the raw time-series, without an access to prior knowledge-based engineered features ([Fig. 2a](#)).

In order to collect the maximum of information from the available finite body of data while being able to assess the classifiers performance, we performed repeated cross-validation of trained models. To validate each model, 20 iterations of random 4-fold splitting were performed, with the loci from the same nucleus always allocated to the same fold to prevent data leakage due to shared morphology features. This allowed a considerable decrease in confidence intervals in sets of estimated model accuracies ([Fig. 2b-d](#)).

Raw features describing motility, morphology and circular organization		
Locus displacement	D	Distance in microns between two consecutively detected locus positions
Nuclear area	A	Nuclear area in square microns
Nuclear perimeter	P	Nuclear perimeter in arbitrary units
Locus distance to nuclear periphery	Dist	Minimal distance from locus to nuclear rim, microns
Change of direction	dTheta	The smaller angle between two consecutive displacements, shows the change in direction of locus movement
Prior knowledge - driven feature engineering		
Mean locus displacement	MD	Mean of time series, calculated for each locus

Table 1

Features extracted from the imaging data collected in this study.

Maximal locus displacement	MaxD	Maximal displacement within time series
Variance of locus displacement	VarD	Variance of locus displacement within time series
Mean nuclear area	MA	Nuclear area, mean of time series, calculated for each nucleus
Mean nuclear perimeter	MP	Nuclear perimeter, mean of time series, calculated for each nucleus
Mean locus distance to nuclear periphery	MDist	Locus distance to periphery, mean of time series, calculated for each locus
Variance of locus distance to nuclear periphery	VarDist	Variance of locus distance to nuclear rim within time series
Relative variance of locus displacement	rVarD	Variance of locus displacement divided by mean displacement (VarD / MD)
Relative variance of locus distance to nuclear periphery	rVarDist	Variance of locus distance to nuclear periphery divided by Mean distance to periphery ($\text{VarDist} / \text{MDist}$)
Total locus displacement	TD	Distance from first to last coordinate of track
Persistence of locus movement	Pers	Total locus displacement divided by track length, shows how directional the movement is.
Fastest locus (True or False)	ifFast	True for the fastest (highest MD) homologous locus in the nucleus, and False for the others
Range of locus distance to nuclear periphery	DistR	Difference between maximal and minimal value of locus distance to nuclear periphery
Total locus radial displacement	TDist	Change in locus distance to nuclear periphery, last position minus first
Central locus (True or False)	ifCentr	True for the homologous locus furthest from nuclear rim (highest Dist) and False for the others

Table 1 (continued)

Trend of change of locus distance to nuclear periphery	sDist	The slope of linear approximation of locus distance to nuclear periphery change
Trend of change of nuclear area	sA	The slope of linear approximation of nuclear area change
Trend of change of nuclear perimeter	sP	The slope of linear approximation of nuclear perimeter change
Mean displacement + 2SD outliers (count)	out2sd	Number of displacements 2*SD longer than MD
Mean displacement + 3SD outliers (count)	out3sd	Number of displacements 3*SD longer than MD

Table 1 (continued)

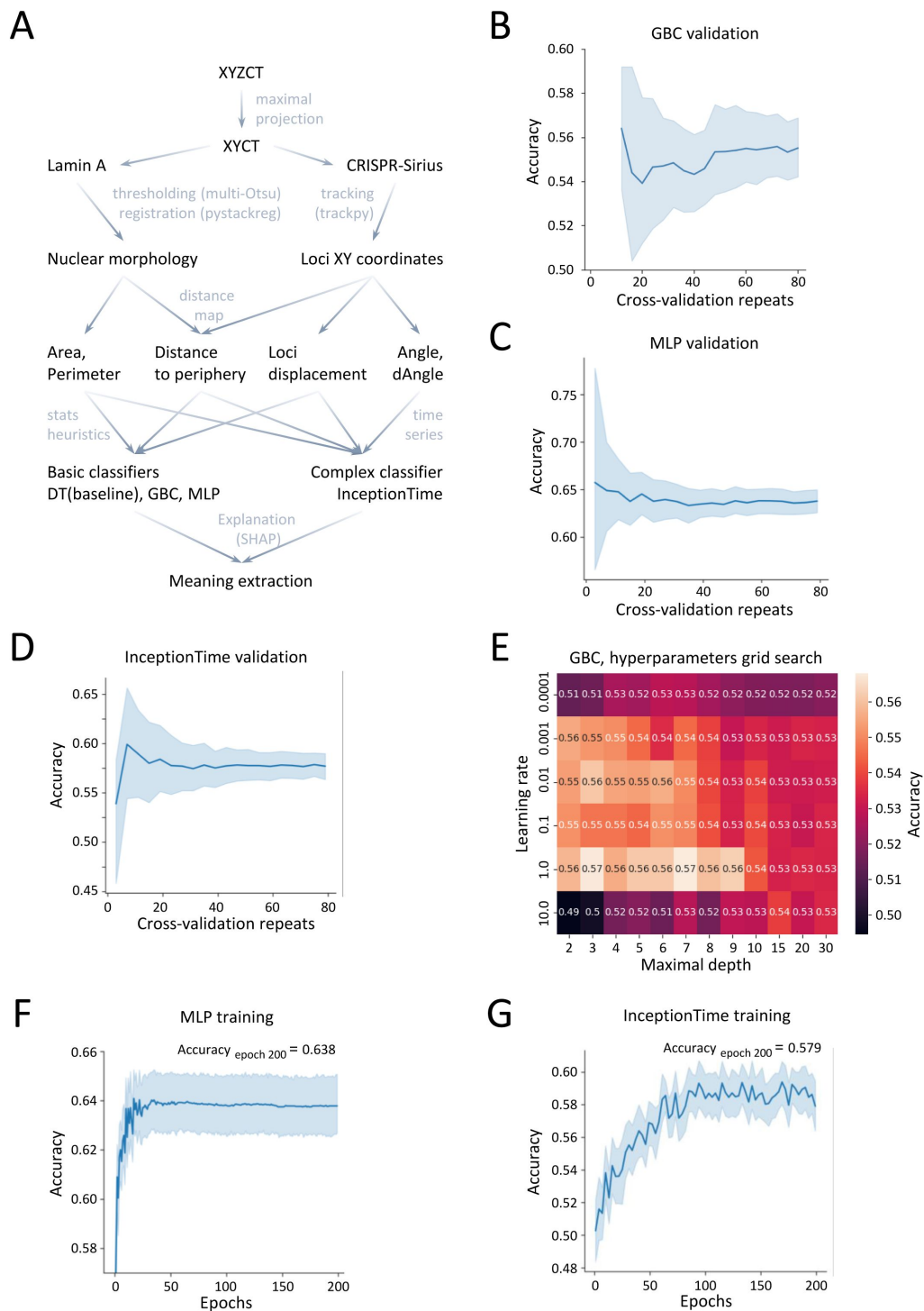


Figure 2

Machine learning-assisted dataset exploration.

a, Information flow from raw pixel intensities to inference and explanations. **(b-d)**, Decrease of accuracy estimation variance in repeated cross-validation of gradient boosting classifier (GBC) **(b)**, multilayer perceptron (MLP) **(c)** and InceptionTime **(d)**. Shaded area shows 95% confidence interval. **e**, Hyperparameter tuning for GBC. **f**, The accuracy of multilayer perceptron converges after approximately 50 epochs of training at validation accuracy of 0.64; **g**, InceptionTime demonstrated performance comparable to GBC, but without access to manually engineered features. **(f,g)** Shaded area shows 95% confidence interval.

Next, having only minor class imbalance (Zero Rate = 0.52), we used one level decision tree that basically thresholds one feature at a time, to estimate baseline performance, which reached an accuracy of 0.54 using the same repeated cross-validation strategy as the rest of classifiers. Remarkably, gradient boosting classifier and multilayer perceptron showed higher mean accuracy 0.57 and 0.64 (**Fig. 2e,f**), indicating the ability of the more complex models to gain additional information by abstracting from the feature interactions and non-linearity. After training on the same dataset, InceptionTime demonstrated accuracy of 0.58 (**Fig. 2g**), exceeding the baseline performance at a level comparable to GBC. Noteworthy, this level was achieved without an access to manually engineered features.

Model decision explanation by SHAP

In order to extract biological insights from the dataset, the rules used by the machine learning model for inference can be extracted in a process called explanation. The basic way to draw the conclusions about the dataset from model performance would be the assessment of feature importances. However, these techniques often return only absolute importance values, do not allow local explanation, and do not specify the interaction component of the features (Saarela and Jauhiainen, 2021). In contrast, a game theory-based approach named SHAP (SHapley Additive exPlanations (Lundberg and Lee, 2017)) returns the locally relevant additive feature impact values, showing whether the feature contributes to or against a particular class and to what extent.

We used SHAP to explain the classifiers' decisions and to assess feature interactions for exploratory purposes (**Fig. 3a-d**). Several features with continuous, non-discrete values, such as mean nuclear area (*MA*) and trend of its change (*sA*), demonstrated high importance and evident correlation between the feature value and its SHAP value in all trained classifiers (**Fig. 3a,b,f**). Interestingly, correlation between speed (see *D* and *MD*) and its impact on model decision was clear in InceptionTime explanation only; the same is true for distance to edge (*Dist* and *MDist*) (**Fig. 3f**). The features with discrete values describing outliers from displacement length distribution (*out3sd* and *out2sd*), also showed clear value/SHAP value correlation in GBC and, more prominently, in MLP (**Fig. 3a-b**), while not used for InceptionTime training. Binary features subclassing the homologous chromosomes based on speed or radial position (*ifFast* and *ifCentr*), did not affect GBC, but demonstrated an impact on MLP decisions that is comparable to other features including morphological (**Fig. 3b**). In accordance with this, discrete and binary features grew in importance in MLP explanation compared to GBC (**Fig. 3e**).

Since chromosome 1 in U2OS cells predominantly has two homologs in each nucleus, information gain from *ifFast* and *ifCentr* cannot be explained by the impact of the feature value itself, but rather implies the influence of these features on other features' SHAP values. Fortunately, SHAP enables estimation of interaction components separated from direct impact of the feature value itself (Molnar, 2019). Indeed, the *ifFast* and *ifCentr* did not have any noteworthy hits on GBC interactions map, while showed strong interaction with each other on MLP (**Fig 3c,d**). MLP feature interactions map in general seemed less noisy, highlighting also the interactions between speed and morphology and total locus displacement and maximal displacement within the timelapse (**Fig 3c,d**). Altogether, the SHAP explanations and the interactions derived from SHAP analysis allowed us to better navigate in the feature space of the dataset, narrowing further analysis to the interpretation of important patterns, as described below.

Nuclear morphology during serum starvation

The explanation of models trained on chromosome 1 tracking data to distinguish high serum condition from serum starvation showed a significant impact of morphological features on classifier decision. Nuclear area (*MA*) and the trend of its change (*sA*) demonstrated highest importance as indicated by SHAP analysis of both GBC and MLP (**Fig. 3e**). Nuclear area also had highest mean absolute SHAP value in InceptionTime predictions (**Fig. 3f**). Indeed, a change in

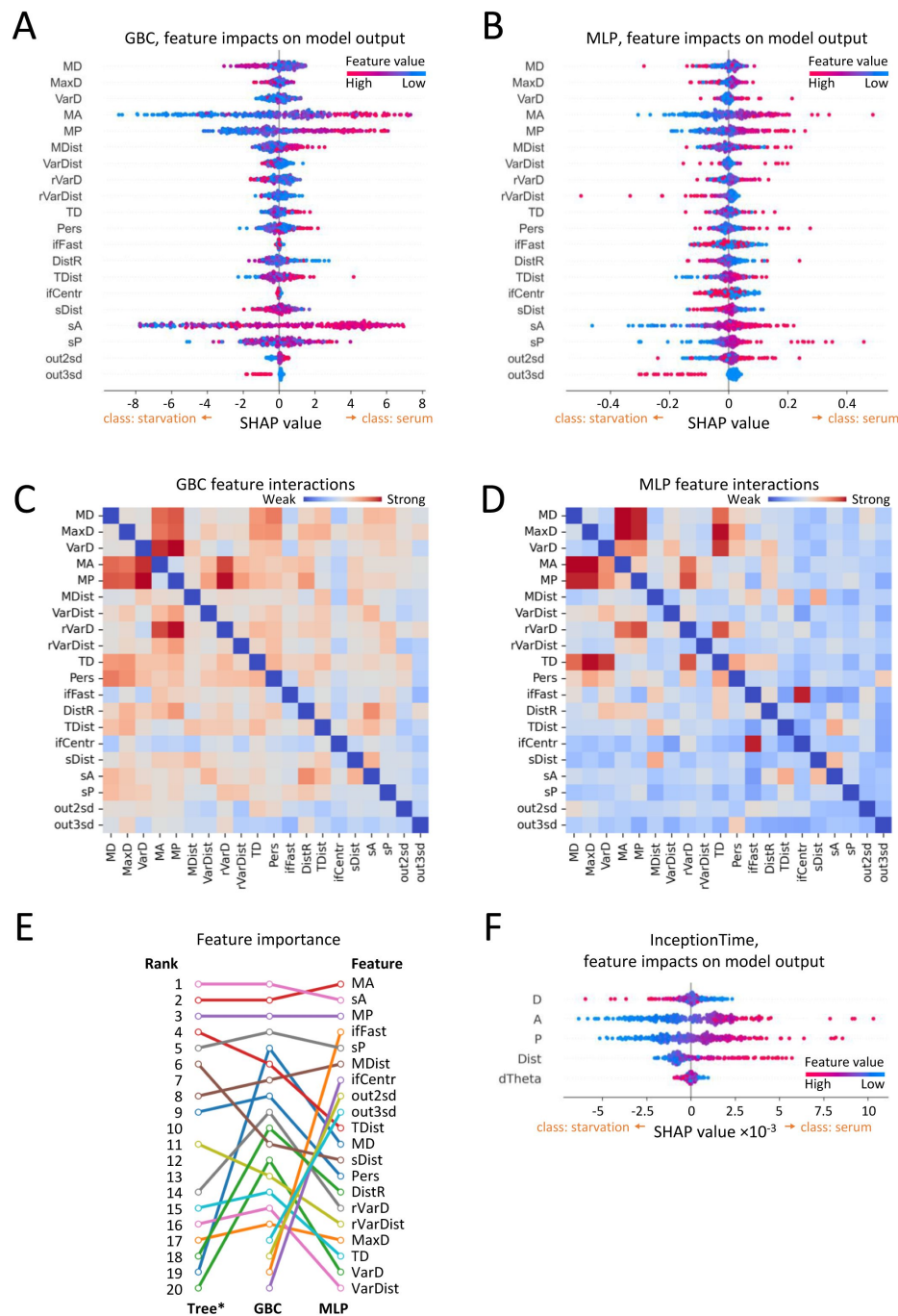


Figure 3

Explaining machine learning models' decision using Shapley additive explanations (SHAP).

a,b Global summary of SHAP showing feature impacts towards one of the classes, high serum or starvation, for gradient boosting classifier (GBC) (**a**) and multilayer perceptron (MLP) (**b**). Feature values are color-mapped from blue (low) to red (high). **c,d** Heatmap of feature interactions depicting the components of synergistic effect of features, after accounting for the direct impact. Interactions from GBC seem to be more uniform and show less usage of binary and discrete features (**c**), compared to the interactions from MLP (**d**). **e**, Change in feature importance rank with the growth of model complexity from the baseline single level decision tree (designated as Tree) to GBC and MLP. Binary and discrete features are not shown for the Tree set, as the model is trained on a single feature at a time to estimate baseline performance, thus, feature interactions are not used by definition. **f**, Global summary of SHAP showing feature impacts towards class serum or starvation, InceptionTime.

the skewness of nuclear area distribution was observed under starvation (**Fig. 4a**) with the detected decrease in the mean value of approximately 50 squared microns, which represents more than 10% of the average nuclear area (**Fig. 4b**, **Supplementary Figure 4a**). Importantly, this change alone cannot explain the radial rearrangement of chromosome 1, as we did not detect any change in telomere distance to the nuclear edge (**Fig. 1f**). The nuclear area also becomes more dynamic under serum starvation (**Fig. 4c**), and the changes in both morphological features were detectable already within the first 20 min after the medium change (**Fig. 4b,c**). Notably, nuclei with an area below 250 squared microns rarely demonstrated further decrease in their area (**Fig. 4d**), perhaps due to the area decrease occurring prior to timelapse recording. Since the length of the nuclear perimeter (MP), similar to the area (MA), decreased in starvation (**Supplementary Figure 4b**), we wondered whether this shrinkage led to a change in circularity, similarly as reported for apoptotic cells (Toné *et al.*, 2007; Helmy and Abdel-Azim, 2012). In our dataset, the changes in both circularity and the correlation between area and perimeter were negligible (**Fig. 4e**, **Supplementary Figure 4c**), confirming the different nature of serum starvation-driven changes in nuclear morphology compared to apoptosis.

Further, the SHAP explanation of both GBC and MLP revealed interactions between speed-related features, such as mean (MD) and maximal (MaxD) locus displacement and morphological features, namely MA and MP (**Fig. 3c** and more prominent in **Fig. 3d**). Upon closer examination, we noticed that high mean displacement values shifted the area's impact on model decisions towards SHAP values around zero (**Fig. 4f**). This was also true for GBC and InceptionTime explanation (**Supplementary Figure 4d-f**), thus showing that the predictive value generated by nuclear area is rather contextual, e.g. its importance is high in a subgroup of low motility loci. In summary, our results demonstrate serum starvation induces a global change in nuclear morphology.

Unique characteristics of chromosome

1 movement under starvation

To achieve the most comprehensive integration of morphology data and chromosome 1 motility, we next augmented the basic raw displacement data with prior knowledge-based engineered features. Initially, no obvious change in the speed of chromosome 1, as measured by mean displacement (MD), was observed under starvation (**Fig. 5a**). However, telomeres seemed to display a trend of decreased motility (**Fig. 5b**), which could be explained by the decrease in nuclear volume (**Fig. 4b**), possibly resulting in increased chromatin density. At the same time, the total distance traveled by chromosome 1 loci within the timelapse (TD) grew in parallel with speed (**Fig. 5a,c**), while no growth in total distance was observed for the telomeres (**Fig. 5d**). Thus, chromosome 1 loci seem to retain their motility level, while telomeres, on average, seem to slow down under starvation. Interestingly, while both nuclear morphology (**Fig. 4b,c**) and telomere dynamics (**Fig. 5b**) responded already during the first 20 minutes of serum starvation, the changes for chromosome 1 were observed later (**Fig. 5a,c**), further highlighting the specific response of chromosome 1 loci for this particular signal.

To further characterize chromosome 1 dynamics, we next focused on persistence, calculated as the ratio of total distance traveled (TD) to the trajectory length ($MD \cdot n_{frames}$). The initial assumption, based on FISH-based studies, that long-range persistent movement would be detectable under starvation did not hold, as persistence did not correlate with persistence SHAP values (**Fig. 3a,b**, **Fig. 5e**). At the same time, SHAP values for total distance (TD) depended on loci speed in a non-linear way, with low-speed loci ($TD < 0.2$) having close to zero SHAP value (**Fig. 5f**). In the high TD range, generally slow-moving loci are associated with negative SHAP values (**Fig. 5f**), thereby influencing the model's decision towards the 'starvation' class. This dependence suggests that speed of locus movement and its directionality do interact in a meaningful way that is specific to chromosome 1. However, this interaction cannot be reduced to the simple measure of persistence.

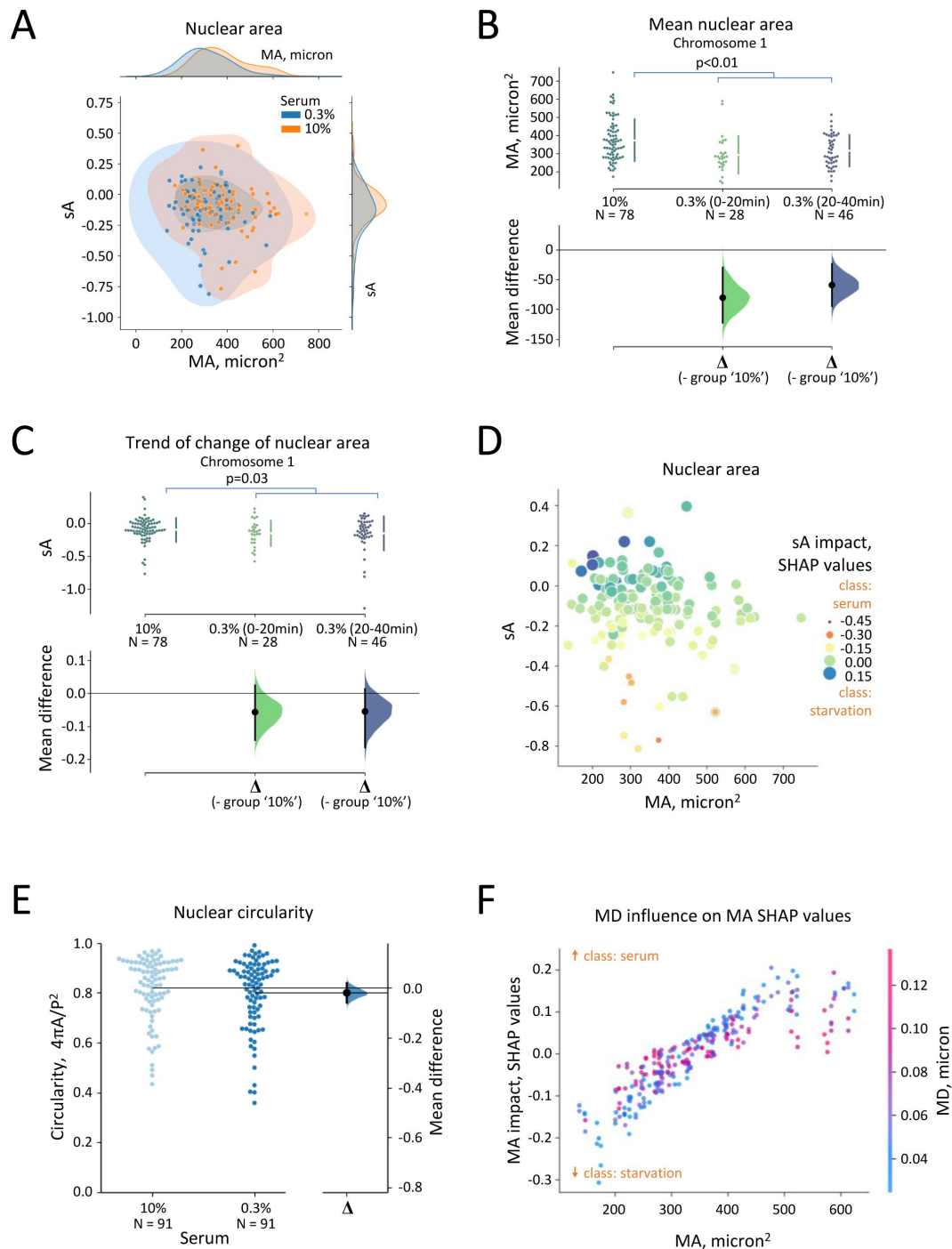


Figure 4.

Changes in nuclear morphology upon serum starvation.

a, Nuclear area (MA) and its dynamics (sA) in high serum medium and under starvation. The marginal distributions show corresponding kernel density estimations. **b-e**, Changes in nuclear morphology, particularly, area (**b**) area dynamics (**c,d**) and circularity. Statistical significance with Mann-Whitney U rank test. (**e**). Each dot in the swarmplot represents an individual nucleus. Panels on the right represent the distributions of bootstrapped delta mean values for datasets. Error bars show 95% confidence intervals. **d**, Impact of nuclear area dynamics (as SHAP values, shown in color) on model decision depending on nuclear area and dynamics of its change. **f**, Locus speed affects Shapley values for nuclear area, which is clear from vertical color gradient, where the locus speed is color-mapped, see legend on the right. MLP SHAP explanation; outliers (less than 2% of data) are not shown for clarity.

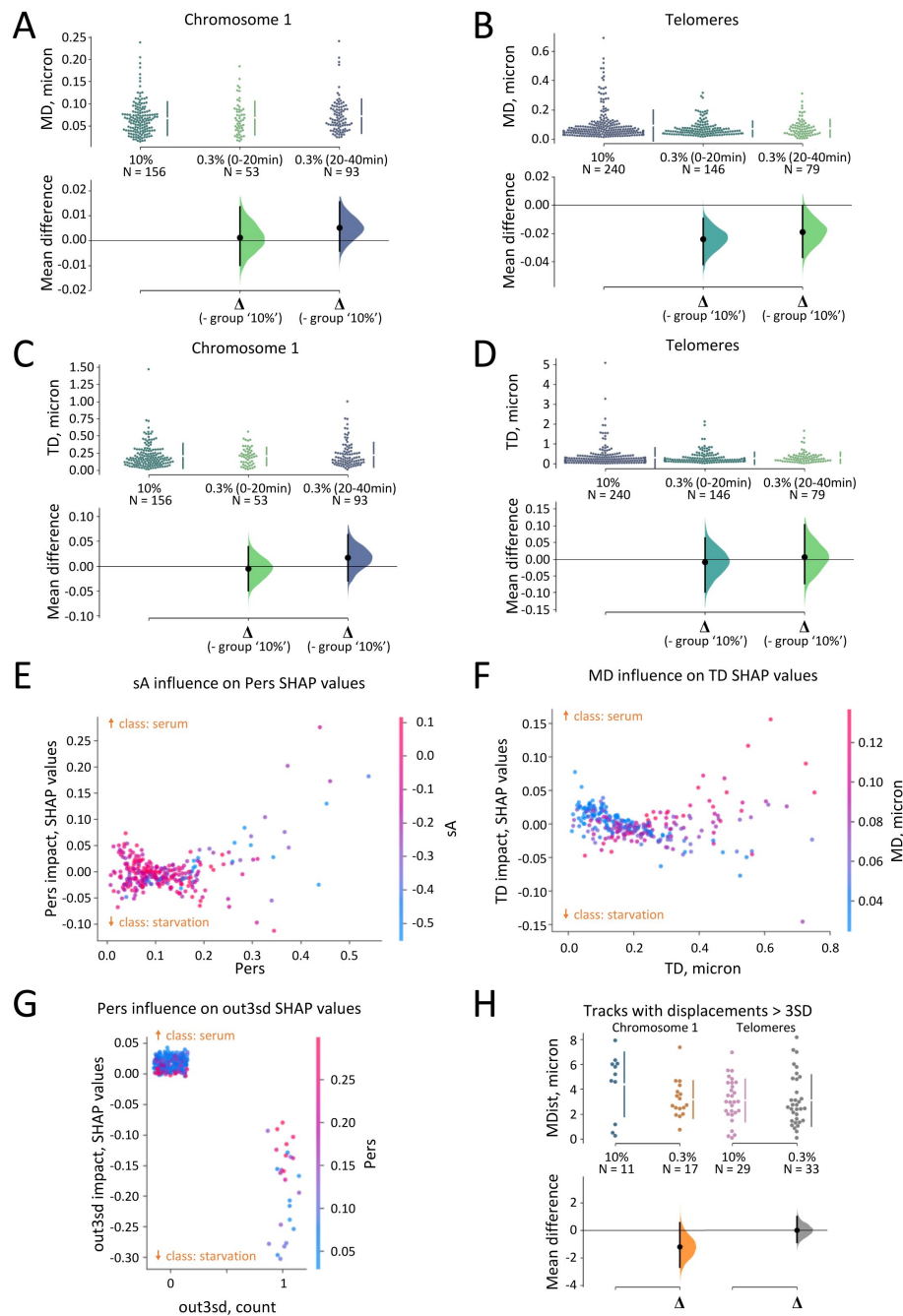


Figure 5.

Specific features of chromosome 1 dynamics.

a-d, Changes in loci motility as measured by mean displacement (MD) and total distance traveled (TD). Mean loci displacement did not change for chromosome 1 under starvation (**a**), but showed a decrease in telomeres (**b**). Total distance traveled by a locus for chromosome 1 grew insignificantly, in a very similar manner to its mean displacement (**c**), while for telomeres stays stable despite the decrease in speed (**d**). Each dot in the swarmplot represents an individual nucleus. Panels on the bottom represent the distributions of bootstrapped delta mean values for datasets. Error bars show 95% confidence intervals. **e-g**, SHAP analysis of features describing persistence and directionality of loci movement. **e**, dynamics of the change of nuclear area (sA) affects the persistence (Pers) SHAP values. **f**, locus mean displacement (MD) affects the SHAP values calculated for total traveled distance (TD). **g**, Persistence (Pers) is negatively correlated with SHAP values of feature which detects long spontaneous displacements (out3sd). **h**, Long spontaneous displacements (out3sd = 1) of chromosome 1 loci are more likely to occur in short distances from nuclear periphery under serum starvation.

Remarkably, the presence of unusually long ($MD+3\sigma$) displacements in chromosome 1 track (feature named *out3sd*), demonstrated a considerable and uniform impact on MLP decision, directed towards starvation class (Fig. 3a,b, Fig. 5g). The SHAP values of *out3sd* were negatively correlated with persistence (Fig. 5g), suggesting that *out3sd*-outliers do not just follow the slow unidirectional movement. Instead, the loci seem to undergo random walk with occasional fast relocalizations. Moreover, under starvation, these long displacements for chromosome 1 loci, but not for telomeres, were detected with higher probability in the proximity of the nuclear envelope (Fig. 5h). Thus, the ability to maintain dynamics in the smaller nuclei (Fig. 5a) alongside intermittent ‘jumps’ (Fig. 5g) especially near the nuclear envelope (Fig. 5h) appear to be hallmarks of chromosome 1 dynamics in a low serum environment.

Radial organization and differential behavior of homologous chromosomes

Being intrigued by the higher occurrence of long displacement outliers close to the nuclear periphery, we analyzed in more detail the set of features related to radial localization of loci, namely *DistR* and *TDist*. *DistR* is the absolute difference between the initial and final radial positions within the timelapse, which does not show the direction, towards the center or the periphery, of the movement. *DistR* exhibited a slight increase in starvation for chromosome 1, while demonstrating a pronounced increase for telomeres (Fig. 6b). In contrast, *TDist* is the total radial distance traveled by a locus, and shows the direction of relocalization. For chromosome 1, the distribution of raw values for *TDist* skewed downwards in serum starvation (Fig. 6c), supporting the relocalization towards the nuclear periphery. The distribution for change of *TDist* in telomeres was much wider, with the mean value closer to zero, while raw values for telomeres were distributed symmetrically (Fig. 6c). Thus, also these features support the chromosome-specificity of radial reorganization.

In line with the observations on radial reorganization (Fig. 1f, Fig. 6c), SHAP dependence plot of the ‘static’ distance to edge, *MDist*, in GBC showed positive correlation between *MDist* and its SHAP (Fig. 3a). While MLP explanation did not show similar clear correlation (Fig. 3b), its butterfly-like shape had a narrowing distribution of SHAP-values at around 3-4 microns from the edge (Fig. 6d). This narrow distribution indicates an area of less predictive power and alludes to the division of the dataset to two separate subclasses - peripheral and central. On the *MDist* SHAP dependence plot based on MLP decisions (Fig. 6d), high total displacement (*TD*) in peripheral subclass pushes the prediction towards starvation class, in line with higher chance to see the *out3sd* outliers in the same region (Fig. 5h). These observations suggest the relevance of subgrouping the chromosome 1 homologs based on radial localization.

To ascertain the influence of radial localization on the differential behavior of chromosome 1 homologs, the loci were grouped based on two factors: speed (*ifFast*) and proximity to the periphery (*ifCentr*). In a primary screening, locus speed (*MD*), total traveled distance (*TD*) and total radial displacement (*TDist*) were correlated between homologs regardless of serum starvation status (Fig. 6e-g). Significant positive correlation between homologs’ speed (Fig. 6e) and total traveled distance (Fig. 6f) was detected, likely showing that those features largely depend on the general degree of constraint in the particular nucleus. If the radial reorganization would be caused by non-specific deformation of the nucleus, the total radial distance traveled by a locus (*TDist*) should be correlated for both homologs. However, our observation shows no correlation (Fig. 6g), indicating that general structural distortion does not affect radial reorganization. This suggests that certain chromosome-specific effects, which may independently affect homologs, could orchestrate radial reorganization during serum starvation.

Curiously, the MLP SHAP explanation highlighted the interaction between the features generating homolog subgroups, *ifFast* and *ifCentr* (Fig. 3d). To further examine this finding, we investigated how the *ifFast* label is distributed between the central and peripheral homologs in the different

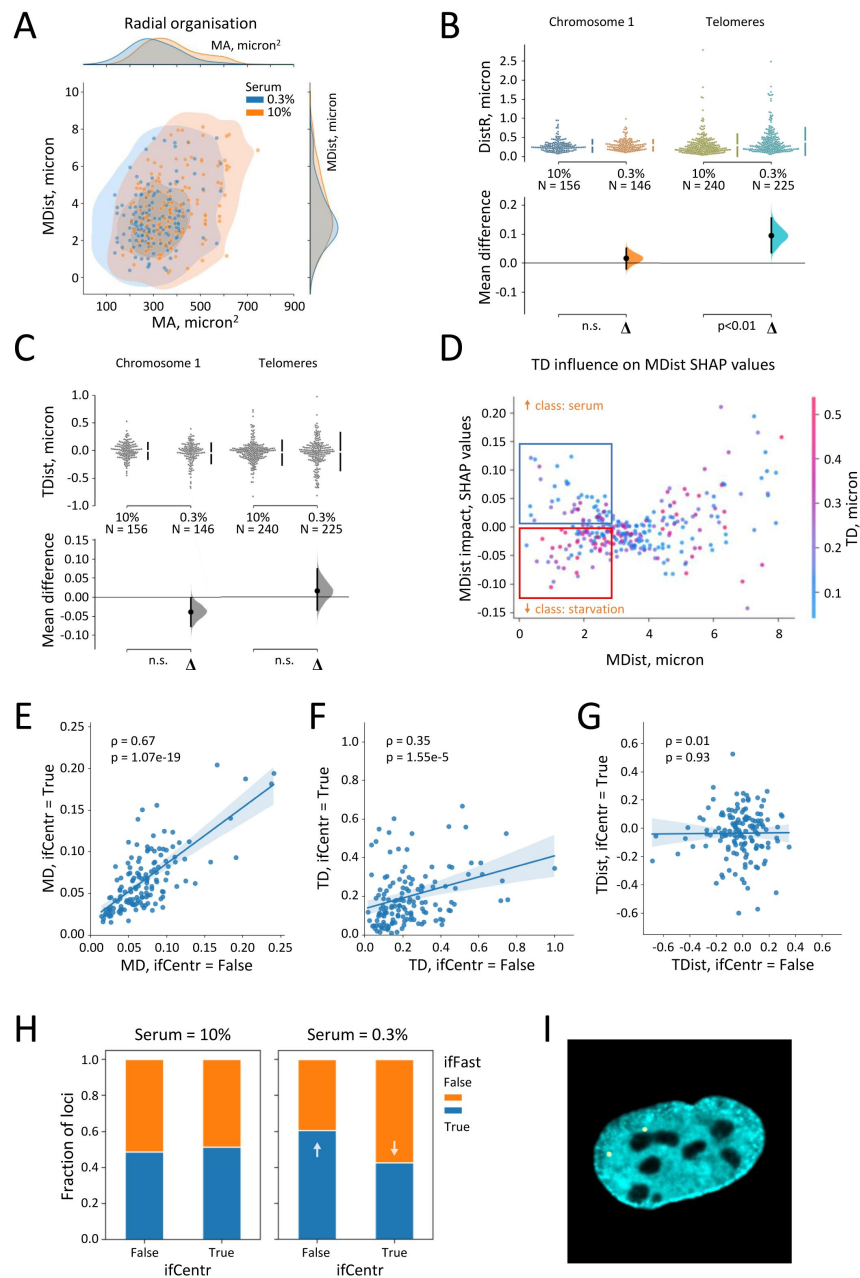


Figure 6.

Distinct dynamic behavior of chromosome 1 homologs upon serum starvation.

a-d, Chromosome 1 radial organization as explored using time series data analysis. **a**, Nuclear area (MA) and distance to nuclear edge (MDist) in high serum medium and under starvation. The marginal distributions show corresponding kernel density estimations. **b,c** Time-lapse-derived data confirm chromatin radial reorganization under serum starvation. The dynamic range of radial movement changed insignificantly for chromosome 1, but showed substantial growth in telomeres (**b**). Total length of radial relocalization (TDist), showing not only scale but also the direction of relocalization, decreased in chromosome 1 but had only insignificant change in telomeres (**c**) Statistical significance (**b,c**) with Mann-Whitney U rank test. **d**, Total distance traveled by locus (TD) affects distance to edge (MDist) SHAP values. **e-i**, Differential behavior of Chromosome 1 homologs. **e-g**, Correlation of mean loci displacement, MD (**e**), total distance traveled by locus, TD (**f**), and total radial displacement, TDist (**g**) between chromosome 1 homologs. **h**, Relationship between categorical features subgrouping chromosome 1 homologs based on speed (ifFast) and radial position (ifCentr). **i**, Representative image of chromosome 1 loci (in yellow), where one of the homologs is associated with a nucleoli-like subcompartment. The LaminaA channel is shown in cyan.

conditions. In the high serum condition, *ifFast* and *ifCentr* are well-balanced, indicating that the peripheral locus has an equal probability of being faster or slower compared to its homolog (**Fig. 6h**, left). However, under starvation, there is a shift in this distribution, associating peripheral loci more frequently with the *ifFast* label (**Fig. 6h**, right). This indirectly supports (i) the findings from the MDist SHAP analysis, which showed that high total displacement (*TD*) is not predictive in central (*MDist* > 3 microns) loci (**Fig. 6d**) and (ii) a higher likelihood of detecting fast spontaneous movements closer to the periphery under starvation (**Fig. 5h**). Overall, this characterization indicates that peripheral loci are more mobile and responsive to stimuli. Intuitively, this may be related to the frequent detection of one of the homologs, often the central one, associated with a subcompartment resembling nucleoli (**Fig. 6i**). Thus, in serum starvation, chromosome 1 homologs display differential behavior, with peripheral loci demonstrating greater mobility and responsiveness to stimuli.

Discussion

Chromatin is dynamic at all length scales, and methods for sequence-specific labeling of chromatin loci are opening new avenues for studying the underlying molecular mechanisms with live imaging techniques. As the dynamics of a specific locus are influenced by the overall chromatin organization, proximity to nuclear landmarks as well as the general nuclear morphology, extracting the relevant patterns from multidimensional imaging data becomes a challenge. Here we have combined CRISPR-Sirius-based labeling of specific chromatin loci for live imaging with machine-learning assisted dataset exploration followed by model explanation with SHAP. This approach uncovered the specific features of chromosome 1 loci dynamics during serum starvation and is widely applicable to dissect mechanistic insights from many different types of experimental data.

Given that ML models vary in efficiency and might highlight different aspects of the dataset under study, a comprehensive exploratory analysis normally includes testing a set of models (Xu and Jackson, 2019; Deng *et al.*, 2021). In our hands, while all the trained models were superior to the baseline (single-level decision tree), MLP, trained on the prior knowledge-based engineered features aggregated over time axis, had the highest accuracy, presumably extracting the most information from the available dataset (**Fig. 2e,f,g**). Despite drastically different architecture, in many cases the trained models generated notably similar patterns of feature importances. For example, locus distance to nuclear periphery was positively correlated with prediction of class ‘serum’ by both GBC and InceptionTime (*MDist* and *Dist*, **Fig. 3a,f**). Another example includes intricate interaction between nuclear morphology and locus speed. For all the used models, a large nuclear area was associated with class ‘serum’, which is evident from positive correlation between area and area SHAP values (**Supplementary Figure 4d-f**). At the same time, for the subgroup of fast loci (high *D*, *MD*) this correlation was lower, while slow loci were characterized by highest absolute SHAP values. In the GBC explanation, this pattern was least clear, but still recognizable; InceptionTime showed SHAP values distribution similar to MLP, with even stronger correlation (**Supplementary Figure 4f**). Overall, this similarity suggests that all the models were able to create the aligned abstractions, accurately approximating the underlying data.

On the other hand, despite general trends being captured by all tested models, certain patterns were specifically highlighted by some of them. Notably, displacement length (*D*) clearly correlated with the starvation class prediction only for InceptionTime. Since this model preprocessed raw time-resolved data by ‘scanning’ it with one-dimensional convolutions, one possible explanation for this observation is that InceptionTime modules effectively extracted specific events or patterns from the time series, analogous to how *out3sd* events were manually extracted to train GBC.

Features describing nuclear morphology showed high importance for decisions of all tested classifiers (**Fig. 3e**), highlighting the profound impact of serum starvation on the cell. Serum starvation is a signal for the cells to become quiescent and exit the cell cycle. It thus affects numerous signaling pathways, and genomic functions from epigenetics to gene expression (Pirkmajer and Chibalin, 2011). It is important to note that here we focused on the very immediate response to starvation and monitor nuclear and chromatin response only during the first hour after changing the media to low serum. The functional implications of chromosome repositioning upon serum starvation are not obvious, but it is tempting to speculate that repositioning of chromosomes to new nuclear environments optimizes their function in the changing environment. Previous studies have shown that almost half of the chromosomes displayed altered radial localization in proliferating compared to quiescent or serum starved cells (Mehta *et al.*, 2010). Indeed, we found that although telomeres in general became less mobile upon serum starvation (**Fig. 5b**), perhaps due to increased chromatin density upon decreased nuclear volume (**Fig. 4b**), they displayed increased radial movement without preferential orientation towards nuclear center or periphery (**Fig. 6c**). This may reflect the overall radial movement of several chromosomes, which could hinder the movement in the lateral dimension. Previous studies with CRISPR-Sirius had indicated cell-cycle dependent chromatin dynamics, with the chromatin movement increasing from early to late G1 phase of the cell cycle, and then decreasing during the S phase (Ma *et al.*, 2019). Here we have restricted our analysis to G1-early S phase of the cell cycle, since we analyzed only cells displaying two “dots” of chromosome 1 loci. In the future, it would be interesting to analyze, whether the cell cycle state influences the chromatin response to serum withdrawal. Nevertheless, the fact that we detected radial reorganization of chromosome 1 loci also when analyzing the same cell in the presence and absence of serum (**Fig. 1g**), indicates that the cell cycle stage is not sufficient to explain the changes in chromatin dynamics.

The overall change in nuclear morphology (**Fig. 4a-d**) and chromatin dynamics (**Fig. 5b**) do not explain the radial relocation of chromosome 1 loci (**Fig. 1f,g**) in response to serum starvation. Instead, chromosome 1 loci display distinct dynamic behavior, which is characterized by maintaining the dynamics (**Fig. 5a**) in the otherwise constrained environment (**Fig. 5b**), and “long-jumps” (**Fig. 5g**), especially near the nuclear periphery (**Fig. 5h**). One example demonstrating the power of SHAP is the interpretation of persistence in locus movement. Since the FISH-based experiments had suggested a role for an actomyosin motor in chromosome repositioning (Mehta *et al.*, 2010), we expected to see evidence of active, directed movement upon serum starvation, similarly as reported, for example, for heat-shock induced repositioning of the HSP70 transgene (Khanna *et al.*, 2014) or movement of heterochromatin breaks in fly cells (Caridi *et al.*, 2018). However, there was no noticeable gain in predictive power from persistence, calculated as $TD/MD \cdot n_{frames}$ in any of the trained models. Nevertheless, TD had a pronounced interaction component in its impact on the model’s decision, highlighting its interaction with speed-related movement characteristics (**Fig. 3d**). Further analysis identified a subgroup of loci that are characterized by high total displacement but low mean displacement under serum starved conditions (**Fig. 5f**), and suggested an association with another feature, namely *out3sd*, which indicates the presence of long displacement outliers within the captured timelapse. In the future, improving the temporal resolution of imaging could provide valuable insights into the nature of the observed fast intermittent movement and potentially prove the role of active transport, perhaps mediated by a nuclear actomyosin motor. Another intriguing question is how the directionality of relocation is established for specific chromatin loci. It would, for example, be interesting to analyze the dynamics of further chromatin loci, such as those moving towards the center of the nucleus.

Majority of the reports on radial chromosome organization neglect the difference in behavior between homologs, operating either with averaged fluorescent signal measured within a radial mask (Mehta *et al.*, 2010), mean centroid position (Mehta *et al.*, 2021), or working with haploid cell lines like HAP1 (Girelli *et al.*, 2020). However, there is a body of data suggesting that

a considerable number of genes, apart from inactivated X chromosome, exhibit monoallelic expression (Gimelbrant *et al.*, 2007 [DOI](#)). Based on these data and measurements of inter-homologue distances, it has been suggested that homologous chromosome territories are positioned in a non-random manner to allow independent regulation, potentially involving the participation of active mechanisms for functional interchromosomal interactions (Heride *et al.*, 2010 [DOI](#)). In this study we report the differential response of homologous loci to serum starvation. Peripheral loci demonstrated longer total displacement when compared to central ones (*TD*, **Fig. 6f** [DOI](#)), and total radial relocation, *TDist*, did not correlate between homologs (**Fig. 6g** [DOI](#)). Moreover, the proportion of faster peripheral loci increased under starvation (**Fig. 6h** [DOI](#)). One explanation to the differential behavior of chromosomes may be anchorage to specific nuclear substructures. Indeed, we often observed one of the homologs in association with structures resembling nucleoli (**Fig. 6i** [DOI](#)), which represent a heterochromatin-associated subcompartment (Bizhanova and Kaufman, 2020 [DOI](#)) similarly as the nuclear periphery. Thus, starvation-driven change in chromatin dynamics can be specific not only to chromosomes, but also to specific homologs.

In general, the explorative approach proposed in this study is conceptually aligned with diffusional fingerprinting as described by Pinholt *et al.* (2021) [DOI](#). While both techniques share the idea of using the enriched feature space to analyze particle tracking data, there are several key distinctions. By design, diffusional fingerprinting employs feature engineering primarily focused on particle biophysics, with the intention to provide generalizable model applicable across diverse types of diffusional behavior. In contrast, we designed the feature set to specifically characterize inherently constrained chromatin movement, while also integrating the relevant data on nuclear morphology. Additionally, the two methods differ in their explainability frameworks. While diffusional fingerprinting utilizes linear discriminant analysis to extract mechanistic insights into particle behavior, SHAP, used in our study, enables greater flexibility when used with complex ML models. The observed relationships between radial organization, speed, total displacement, and directionality suggest that within the analyzed feature space, the machine learning models were able to identify patterns that are more informative about the experimental conditions under study than any of manually engineered features based on prior knowledge by itself. We believe that the reported dataset exploration technique offers a powerful tool for effectively mining complex patterns within datasets in both basic biology and biomedicine.

Materials and methods

Plasmids and guide RNA

CRISPR-Sirius system plasmids, namely pHAGE-TO-dCas9 (Addgene ID 75381), pHAGE-EFS-PCP-GFPnls (Addgene ID 121938), pLH-sgRNA-Sirius-8xPP7 (Addgene ID 121940) were kindly provided by Pederson lab (Ma *et al.*, 2018 [DOI](#)). sgRNA was cloned into pLH-sgRNA as described in (Ma *et al.*, 2016 [DOI](#)) using Stbl3 competent cells. sgRNA sequences:

Chromosome 1: AGATGCTCACC and AGGCTGAGGCTA

Chromosome 10: ACAGCGAGCAAG

Chromosome 13: TAAGCATGGACCATTCTTC

Chromosome X: TAGAGTCTGCAT Telomeres: TAGGGTTAGGGTTAGGGTTA

LaminA-mCherry construct was generated by swapping EGFP to mCherry in pEGFP-LaminA vector. Plasmids created for this study are available upon request.

Live cell imaging

For live cell imaging, human U2-OS cells, a kind gift from Lappalainen-lab (University of Helsinki) and regularly tested for mycoplasma contamination, were cultured routinely in DMEM medium supplemented with 10% FBS (GIBCO), 100 Units/ml penicillin, and 0.1 mg/ml streptomycin (Pen-Strep, GIBCO), then trypsinized and transferred to 35mm glass bottom Mattek dish following transfection with Lipofectamine 3000 (Thermofisher) according to manufacturer's protocol. 200 ng of pHAGE-TO-dCas9, 40 ng of pHAGE-EFS-PCP-GFPnls, 1000 ng of pLH-sgRNA-Sirius-8xPP7 and 150 ng of LaminA-mCherry plasmids were used per 35 mm dish. Live cell imaging was performed 24-36 h after transfection in FluoroBrite DMEM (Thermofisher). First, cells were imaged in 10% serum (FBS, GIBCO), and then medium changed to 0.3% serum. Imaging was performed using an Intelligent Imaging Innovations (3i) Marianas Inverted Spinning Disk (Yokogawa CSU-X1 M1 5,000 rpm) microscope, equipped with Andor Neo sCMOS camera and solid-state lasers (488 nm/150 mW, 561 nm/50 mW). Cells were imaged using a 63x/1.2 W C-Apochromat Corr WD=0.28 M27 (Zeiss). Data is from at least three biological replicates, with the number of loci analyzed indicated in the figures. For the experiments where the same cells were imaged before and after the medium change (paired data), XY coordinates were stored for the set of cells and the medium was changed on the stage. The CO2 level and temperature were controlled using the environmental chamber and CO2 mixer with active humidity control (Life Imaging Services). For each cell, a Z-stack of 5 sections was captured, with 1 micron step size, 2 min total timelapse duration, 0.25 frames per second.

Image segmentation, loci tracking and data pre-processing

Image format conversion from proprietary .sld (3i) to .ome.tiff (OME) was done using Bio-Formats plugin (Linkert et al., 2010) before the z-projection as a part of custom-made script in ImageJ, 1.51f (Schneider, 2012 #2194). Next, files and metadata were read in python using *apeer_ometiff_library*. Nuclei were segmented using multi-Otsu and morphology toolbox from *scikit-image*, based on LaminA-mCherry channel, to get nuclear morphology characteristics, namely nuclear area and perimeter. Nuclear circularity was approximated as $4\pi A / P^2$, where **A** and **P** stand for area and perimeter correspondingly. Rigid body registration (*pystackreg*) was used to compensate for the nucleus translation and rotation between frames. Chromosome loci visualized by CRISPR-Sirius were detected and tracked by the *trackpy* (Crocker and Grier, 1996 [DOI](#)), to acquire raw data on loci displacements in between frames. The binary masks of nuclear perimeter and xy coordinates of chromosome loci were used for mapping the loci distance to nuclear rim using *numpy*. Finally, the data collected in tracking experiments was aggregated, cleaned and further used for prior knowledge-based feature engineering mainly using *pandas*, *numpy* and *scipy.stats*. For more details of image analysis please see the code, link provided in the Data Availability section.

Machine learning on tabular data

Pipeline architecture

The core idea behind ML-assisted dataset exploration involves training ML models on tracking data using experimental conditions as labels. The Shapley values-based approach is then employed to explain the models and uncover higher-level descriptive patterns used by the models for classification. An ensemble of gradient boosted decision trees (*scikit-learn*, designated GBC) and a small 400k parameter neural network consisting of seven fully-connected layers (*keras*, designated in text as MLP) were chosen due to robustness, computational lightness and accessibility regardless of operators level of technical knowledge.

Baseline

The classes in the dataset under study were almost perfectly balanced, thus threshold for predictive power set at ZeroR = 0.52. To evaluate the ability of GBC and MLP to infer relatively complex patterns from the available data, we established an additional baseline by measuring the performance of a single-level decision tree using the same evaluation scheme employed for GBC and MLP.

Hyperparameters screen and performance evaluation

A hyperparameter search was conducted for GBC, exploring the range of learning rates and maximal tree depths; for the MLP architecture, the depth and number of neurons in dense layers were manually adjusted for tuning. The optimal configurations were further evaluated using repeated StratifiedGroupKFold cross-validation, with the groups defined by association to a specific nucleus, therefore preventing the data leakage due to shared morphology features. Data preprocessing, grid search and cross-validation were done in *scikit-learn* environment. For InceptionTime, the number of inception modules (affecting the overall model size) and the kernel sizes of the modules' 1D convolution layers (the time window lengths) were altered for hyperparameter tuning.

SHAP explanation

Shapley values-based explanation (Lundberg and Lee, 2017 [↗](#)) was done in parallel with cross-validation. Calculated values were aggregated and accumulated through cross validation repeats, taking advantage of additivity, an intrinsic feature of Shapley values. The approximate feature interaction index values were calculated using the *potential_interactions* function from *shap* utils, with the final interaction matrix (A_{si}) being calculated as entrywise sum of the output from *potential_interactions*, designated A_i and its transposed version (A_i^T), divided by two (scalar multiplication), as in $A_{si} = 0.5(A_i + A_i^T)$

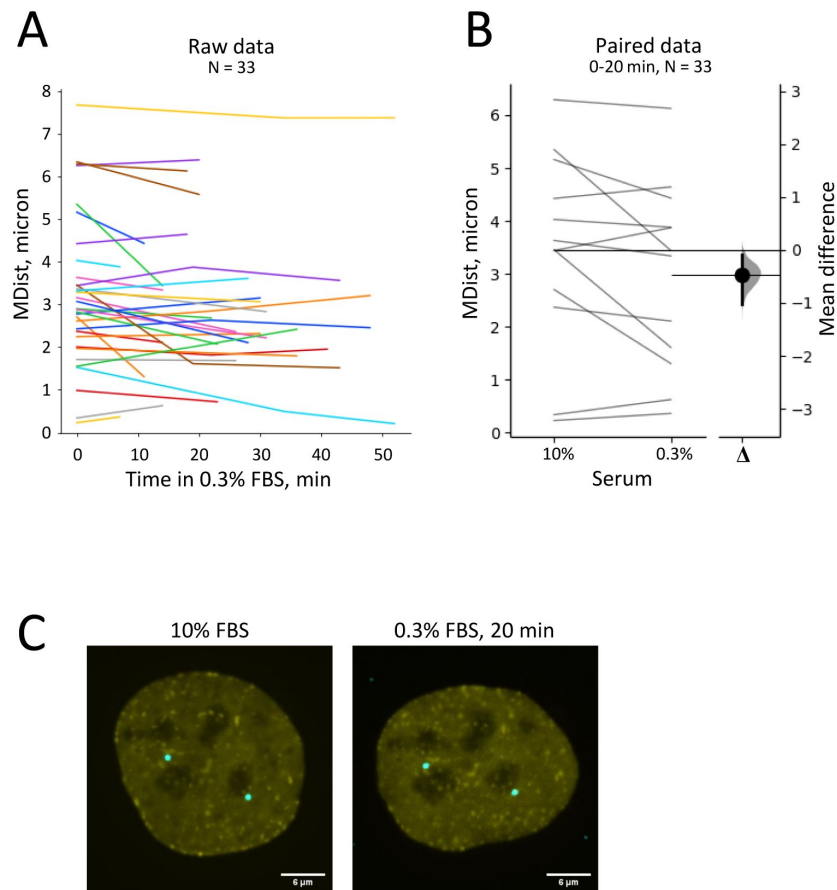
Data visualization

Data was visualized using *matplotlib*, *seaborn* and *dabest* (Ho et al., 2019 [↗](#)). SHAP visualizations were done using a built-in toolbox from the *shap* library. Statistical significance was tested with Mann-Whitney U rank test (independent samples, two-sided) or t-test (paired, two-tailed), as indicated.

Data availability

The code used in this study is available at <https://github.com/redchuk/chromatin-tracking-exploration> [↗](#)

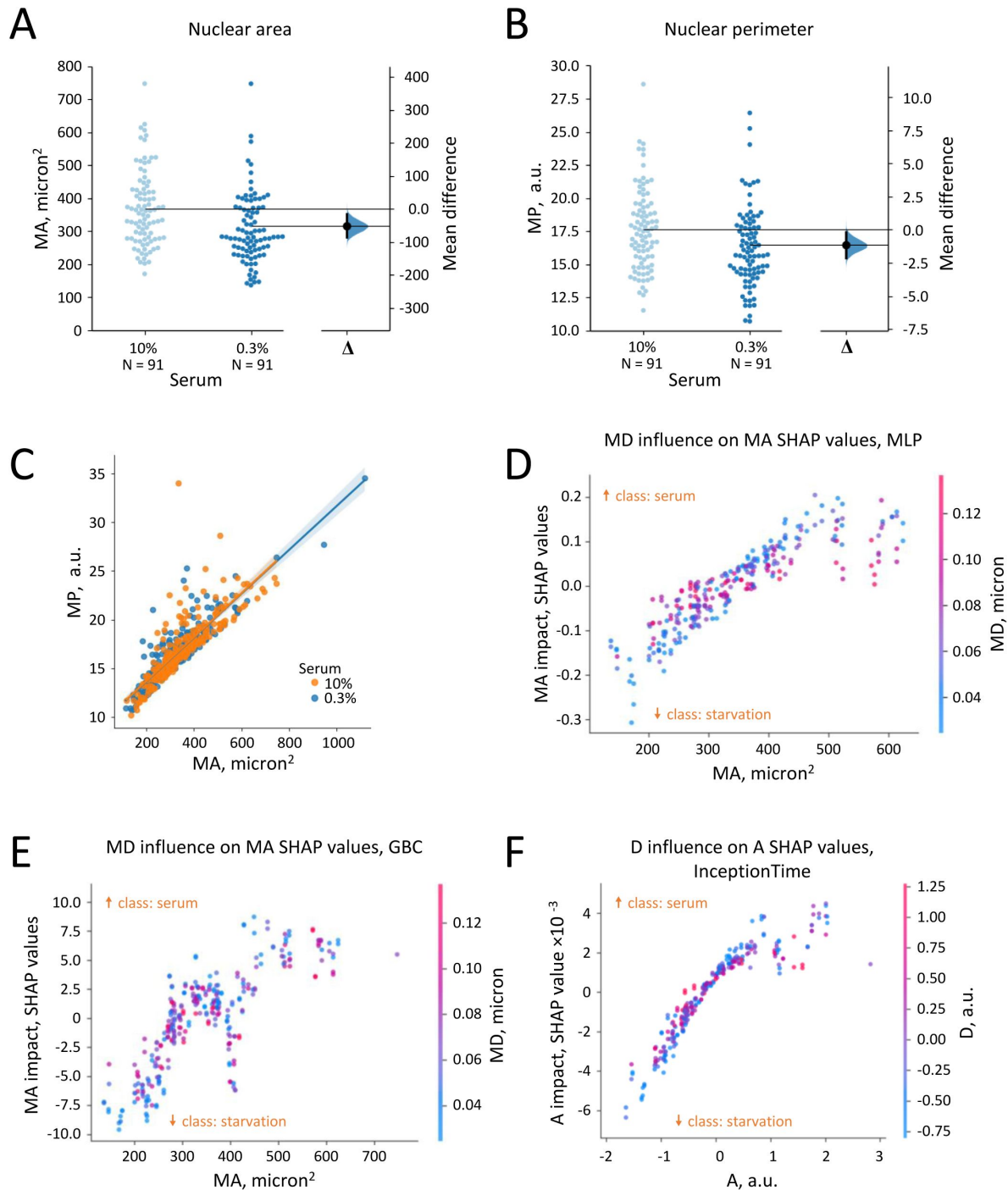
Supplementary Figures



Supplementary figure 1, related to main figure 1.

Mean locus distance to nuclear periphery for chromosome 1 loci under starvation, measured in the same cell before and after medium change.

a, Raw data plotted without time-related subsampling, each line represents a nucleus. **b**, Cells sampled within the first 20 min after medium change). Every line represents the change in minimal distance in particular locus, left panel. The distribution shows bootstrapped delta mean values for paired dataset, right scale. Error bars show 95% confidence intervals. **c**, Representative image of U2OS cells overexpressing Sirius plasmids with guide for chromosome 1 locus, in 10% serum and under starvation. The descriptive statistics for cells imaged before and after the medium change is shown in [Figure 1h](#). Scale bar = 6μm.



Supplementary figure 4, related to main figure 4.

Changes in nuclear morphology under starvation, particularly, area (MA, a) and perimeter (MP, b).

Each dot in the swarmplot represents an individual nucleus. Panels on the right represent the distributions of bootstrapped delta mean values for datasets. Error bars show 95% confidence intervals. **c**, Correlation between nuclear area (MA) and the length of perimeter (MP) in 10% FBS and in starvation (0.3% FBS). Shaded area shows 95% confidence interval. **d-f**, Locus speed as measured by mean locus displacement (MD) affects Shapley values for nuclear area (MA), which is clear from vertical color gradient on SHAP dependence plot, where the locus speed is color-mapped, see the legend on the right. SHAP was used to explain the decisions of MLP (**d**), repeated from Fig 4f for comparison), GBC (**e**), and InceptionTime (**f**) classifiers.

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Additional information

Author contributions

MKV and TR designed the study. ON cloned and tested the sgRNA constructs for CRISPR-Sirius; TR, AP and ON performed the imaging. TR and HJ wrote the code and performed the data analysis. LP helped to design machine learning experiments and performed code review. MKV and TR did the biological interpretation and wrote the manuscript. All authors have seen and approved the manuscript.

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Reviewer #1 (Public review):

Summary:

Redchuk et al. explore the dynamic properties of chromatin upon serum starvation using machine learning approaches. They use CRISPR-tagging to visualize a region on chromosome 1 in human cells and show that in their system, chromosome 1, but not the previously reported chromosomes 10, 13, and X, undergo a change in radial position upon serum starvation. Live cell imaging showed a position change towards the periphery after serum starvation. They then apply a machine learning algorithm for the analysis of the imaging data, which reveals changes in nuclear area during serum starvation and longer displacements of the chromosome 1 locus near the nuclear periphery. Differential behavior of homologues is also reported.

Strengths:

- (1) The study of chromatin dynamics is an interesting and important area of research.
- (2) The use of machine learning approaches to analyze live cell imaging data is timely.
- (3) With serum starvation, the authors use a simple, well-controllable model system.

Weaknesses:

- (1) This study only provides limited new insight into chromatin dynamics.
- (2) It was not immediately evident what the use of machine learning approaches added to this study. It appears that the main conclusions could have been reached by conventional analysis.

(3) There are several specific technical points:

a) It was not clear what the CRISPR-Sirius probes actually labelled. The chromosome 1 sgRNA sequence is provided, but I could not find information as to which region(s) of the chromosome are actually labelled (size, location, etc.).

b) The authors visualize a relatively small region of chromosome 1 but make conclusions regarding the entire chromosome. Additional probes on the same chromosome should be used.

Related to this point, the discussion of why the authors are unable to reproduce the prior findings of relocation of chromosomes 10, 13, and X is not satisfying. It would be worth comparing the FISH-based painting of entire chromosomes, which generated the results suggesting relocation of these chromosomes, with the point-labelling method used here.

c) The study lacks controls. Since in their hands chromosomes 10, 13, and X do not change position, they should be used as a negative control in all experiments demonstrating a shift in the location of chromosome 1.

d) I did not find information about the spatial or temporal resolution of the imaging modality. This is important to assess whether the observed changes in position, relative to time, are meaningful.

e) The authors analyze surprisingly early timepoints (up to 40 minutes) of serum starvation. Would these results look different if longer serum starvation timepoints of several hours were analyzed?

f) The authors can do a better job of explaining what the biological meaning of the various parameters (DistR, TDist, etc.) they measure is.

g) I did not understand the reasoning for the authors' conclusion of differential behavior of homologues. Please explain this better, or ideally use more direct labeling methods that identify the individual homologues.

h) In many figures, statistical analysis of the data is missing, including, but not limited to, Figures 1B, C, G, Figures 4, 5, 6.

i) No information is provided throughout the manuscript as to how many cells were analyzed in each experiment. This should be indicated in every figure legend.

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Reviewer #2 (Public review):

Summary:

The study demonstrates that CRISPR-Sirius provides a powerful approach to investigating chromosome dynamics in living cells during environmental stress. By focusing on serum starvation, the authors show that this process induces global nuclear changes, including a reduction in nuclear area and increased morphological dynamism, while at the same time driving specific reorganization of chromosome 1. Chromosome 1 relocates toward the nuclear periphery and displays distinctive patterns of motion, maintaining overall motility but punctuated by occasional long-distance displacements, particularly near the nuclear envelope. Importantly, the analysis reveals that homologous copies of chromosome 1 do not behave uniformly: peripheral loci become more mobile and responsive to starvation, whereas central homologs remain comparatively stable, often associated with nucleolar

subcompartments. By integrating live imaging with machine learning and explainable AI analysis, the study highlights the complexity of nuclear organization and provides valuable insights into how chromosome-specific and locus-specific responses to stress are orchestrated within the three-dimensional nuclear landscape.

Strengths:

The study uses live-cell imaging to investigate the dynamics of loci during starvation. Live-cell tracking and data interpretation are carried out using machine learning and AI models, which is a major strength.

Weaknesses:

The manuscript is at times difficult to follow, partly because the methodological descriptions are highly specialized, especially for non-expert biologists. In addition, the observations are not tested for a mechanistic basis. Experiments that could provide deeper insights are missing, for example, why chromosome 1 moves, why the peripheral homologue dislocates, or why a "long jump" is observed at the periphery even though the speed of the loci does not change. It is also unclear whether a displacement of 0.5 μm is functionally meaningful.

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